

SYSTEM REQUIREMENTS DOCUMENT (SRD)

Project Name: Cyndy's Sweettooth Online Application

Document Type: System Requirements Specification

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1. INTRODUCTION

1.1 Purpose

This document defines the system-level requirements for the c. It translates the approved business requirements into detailed system specifications that guide development, testing, and deployment.

1.2 Scope

The system will be a web-based application that enables customers to place bakery orders online while allowing the bakery owner and staff to manage products, inventory, and sales data through a centralized system.

2. SYSTEM OVERVIEW

The Cyndy's Sweettooth Online Application will:

- Provide an online platform for customers to browse and order products
- Enable bakery staff to manage and fulfill orders
- Allow the bakery owner to track inventory and sales
- Store and manage operational data securely

The system will be hosted online and accessible via web browsers on desktop and mobile devices.

3. SYSTEM ARCHITECTURE

The system shall consist of:

- **Frontend (User Interface Layer)** – Web interface accessible via browser
- **Application Layer (Business Logic)** – Processes orders, inventory updates, and reporting
- **Database Layer** – Stores customer orders, product data, inventory levels, and reports
- **Hosting Environment** – Cloud-based server infrastructure

4. USER ROLES

The system shall support the following user roles:

1. Customer:

- Browse products
- Place orders
- View order confirmation/status

2. Admin (Bakery Owner):

- Manage products
- View reports
- Monitor inventory
- Manage users

3. Staff:

- View incoming orders
- Update order status

5. FUNCTIONAL REQUIREMENTS

5.1 Customer Module:

FR-01: The system shall allow customers to view available bakery products.

FR-02: The system shall display product name, price, and description.

FR-03: The system shall allow customers to add products to a cart.

FR-04: The system shall validate required fields before order submission.

FR-05: The system shall generate a unique order ID upon successful order placement.

FR-06: The system shall send an order confirmation notification.

FR-07: The system shall allow customers to view order status.

5.2 Order Management Module

FR-08: The system shall allow staff to view incoming orders in real-time.

FR-09: The system shall allow staff to update order status (e.g., Pending, In Progress, Completed).

FR-10: The system shall timestamp all order status updates.

5.3 Inventory Management Module

FR-11: The system shall automatically reduce inventory levels when an order is confirmed.

FR-12: The system shall notify the admin when stock levels fall below a predefined threshold.

FR-13: The system shall allow the admin to manually adjust inventory levels.

5.4 Product Management Module

FR-14: The system shall allow the admin to add new products.

FR-15: The system shall allow the admin to update product details.

FR-16: The system shall allow the admin to deactivate unavailable products.

5.5 Reporting Module

FR-17: The system shall generate daily sales reports.

FR-18: The system shall generate weekly and monthly sales summaries.

FR-19: The system shall display best-selling products.

6. NON-FUNCTIONAL REQUIREMENTS

6.1 Performance

NFR-01: The system shall load pages within 3 seconds under normal usage.

NFR-02: The system shall support at least 50 concurrent users.

6.2 Availability

NFR-03: The system shall be available at least 99% during business hours.

6.3 Security

NFR-04: The system shall require authentication for admin and staff users.

NFR-05: The system shall encrypt sensitive user data.

NFR-06: The system shall implement role-based access control.

6.4 Usability

NFR-07: The system shall provide an intuitive and user-friendly interface.

NFR-08: The ordering process shall not exceed 5 steps.

6.5 Compatibility

NFR-09: The system shall be compatible with modern web browsers (Chrome, Safari, Edge).

NFR-10: The system shall be responsive and accessible on mobile devices.

7. DATA REQUIREMENTS

The system shall store the following data:

- Customer information (Name, Contact Details)
- Order details (Order ID, Products, Quantity, Status, Timestamp)
- Product information (Name, Price, Availability)
- Inventory records
- Sales transaction records

Data shall be backed up daily.

8. ASSUMPTIONS

- Users have internet access.
- The bakery operates from a single location.
- Staff have basic digital literacy.

9. CONSTRAINTS

- Limited development budget.
- Project timeline of approximately 9 weeks.
- Dependence on internet connectivity.

10. ACCEPTANCE CRITERIA

The system shall be considered complete when:

- All functional requirements are implemented and tested.
- Admin and staff users can log in and manage operations successfully.
- Customers can place and track orders without errors.
- Sales and inventory reports generate accurate data.